

Technical Note: Data Inversion Theory for the DMA-APM-CPC Tandem Aerosol Measurement System

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Abstract

This document provides a self-contained description of the mathematical theory and numerical algorithms implemented in the DMA-APM mass distribution inversion tool. It covers the formulation of the forward problem (1D and 2D integral models), the physical justification for the 1D approximation, the original RK4-based APM transfer function simulation, the Chahine–Twomey inversion algorithm with Markowski smoothing, and the Poisson-statistics-based convergence criterion. It also describes the use of the standalone transfer function for planning APM set-points (rotation speed and voltage) for spherical particles and multiply charged sphere clusters.

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1 Objective and problem statement

The objective of this analysis is to estimate the highly resolved mass distribution of aerosol particles that have been pre-classified by a specific electrical mobility diameter using a DMA-APM-CPC tandem measurement system. Specifically, we seek the mass distribution function at the DMA outlet:

$$f(m) = \frac{dN}{dm}, \quad (1)$$

where N is the particle number concentration and m is the particle mass. The observed CPC concentration $n(V)$ at each APM applied voltage V is related to $f(m)$ through a Fredholm integral equation of the first kind, and recovering $f(m)$ from $n(V)$ constitutes an ill-posed inverse problem.

2 Formulation of the forward problem

2.1 The rigorous 2D-integral model

The expected particle number concentration $n(V)$ observed downstream of the APM is governed by a double integral over both mass m and electrical mobility Z_p :

$$n(V) = \int_0^\infty \int_0^\infty \Omega_{\text{APM}}(m, V, Z_p) f_{\text{in}}(m, Z_p) \Omega_{\text{DMA}}(Z_p) dZ_p dm, \quad (2)$$

where $f_{\text{in}}(m, Z_p)$ is the intrinsic 2D mass-mobility distribution of the raw aerosol entering the DMA, $\Omega_{\text{DMA}}(Z_p)$ is the DMA transfer function, and $\Omega_{\text{APM}}(m, V, Z_p)$ is the APM transfer function.

Assuming that f_{in} is approximately constant with respect to Z_p over the narrow DMA transmission window, we define the target distribution $f(m)$ as the mass distribution of the aerosol exiting the DMA:

$$f(m) = f_{\text{in}}(m, Z_p^*) \int_0^\infty \Omega_{\text{DMA}}(Z_p) dZ_p. \quad (3)$$

Using this definition, the forward problem reduces to:

$$n(V) = \int_0^\infty K_{\text{eff}}^{2\text{D}}(V, m) f(m) dm, \quad (4)$$

where the effective 2D kernel is:

$$K_{\text{eff}}^{2\text{D}}(V, m) = \frac{\int_0^\infty \Omega_{\text{APM}}(m, V, Z_p) \Omega_{\text{DMA}}(Z_p) dZ_p}{\int_0^\infty \Omega_{\text{DMA}}(Z_p) dZ_p}. \quad (5)$$

In the implementation, the DMA transfer function is modelled as a triangular function centred at the target mobility Z_p^* with half-width βZ_p^* , where $\beta = Q_{\text{sample}}/Q_{\text{sheath}}$ is the DMA flow ratio. The integral in (5) is evaluated numerically using the rectangle rule with a configurable number of quadrature points.

2.2 The 1D-integral approximation

By approximating the DMA output as a Dirac delta function centred at the target mobility Z_p^* , the effective kernel simplifies to:

$$K^{1\text{D}}(V, m) = \Omega_{\text{APM}}(m, V, Z_p^*), \quad (6)$$

and the forward problem becomes:

$$n(V) \approx \int_0^\infty K^{1D}(V, m) f(m) dm. \quad (7)$$

In both the 1D and 2D models, the inversion algorithm solves for the same target: $f(m)$, the mass distribution of the particles at the DMA outlet.

3 Physical justification: why the 1D model is practically sufficient

In practice, the mass distributions obtained from the 1D and 2D models are virtually identical, even when the DMA resolution is relatively broad (e.g., $\beta = 0.2$). This equivalence arises from the fundamental classification principle of the APM. The central classified mass is:

$$m_c = \frac{eV}{r_c^2 \omega^2 \ln(r_2/r_1)}, \quad (8)$$

where e is the elementary charge, $r_c = (r_1 + r_2)/2$ is the gap centre radius, ω is the angular velocity, and r_1, r_2 are the inner and outer cylinder radii. This expression is completely independent of the particle's electrical mobility Z_p . A variation in Z_p (due to the DMA's finite transmission width) only alters the particle's radial drift velocity inside the APM, slightly affecting its transit time. Consequently, Z_p variations marginally broaden the width of Ω_{APM} but do not shift its central peak. Because the 2D convolution averages these minor, symmetric width changes, the resulting K_{eff}^{2D} is nearly indistinguishable from K^{1D} .

The 1D approximation therefore provides mathematically and physically robust results at approximately $1/N_{Z_p}$ of the computational cost of the 2D model, where N_{Z_p} is the number of Z_p quadrature points.

4 APM transfer function: particle trajectory simulation

4.1 Equation of motion

To compute $\Omega_{APM}(m, V, Z_p)$, this tool employs an original numerical simulation method that directly tracks particle trajectories within the APM gap using the 4th-order Runge–Kutta (RK4) method.

Inside the APM annular gap (inner radius r_1 , outer radius r_2 , centre radius $r_c = (r_1 + r_2)/2$, half-width $\delta = (r_2 - r_1)/2$), a particle of mass m and electrical mobility Z_p experiences centrifugal and electrostatic forces. Under the assumption of a fully developed parabolic laminar flow profile, the ratio of the radial to axial velocity components is (Ehara et al., 1996):

$$\frac{dr}{dz} = \frac{v_r(r)}{v_z(r)} = \frac{\frac{Z_p}{e} \left(m\omega^2 r - \frac{eV}{r \ln(r_2/r_1)} \right)}{\frac{3Q}{4\pi\delta r_c} \left[1 - \left(\frac{r - r_c}{\delta} \right)^2 \right]}, \quad (9)$$

where Q is the volumetric aerosol flow rate. This can be rearranged to:

$$\frac{dr}{dz} = \frac{8\pi\delta r_c}{3Q} \frac{Z_p}{e} \cdot \frac{m\omega^2 r - \frac{eV}{r \ln(r_2/r_1)}}{1 - \left(\frac{r - r_c}{\delta} \right)^2}. \quad (10)$$

4.2 RK4 integration and transmission efficiency

The trajectory equation (10) is integrated along the axial coordinate z from $z = 0$ to $z = L$ (the electrode length) using the classical 4th-order Runge–Kutta method with a fixed step size Δz :

$$k_1 = \Delta z \cdot g(r_n), \quad (11)$$

$$k_2 = \Delta z \cdot g(r_n + k_1/2), \quad (12)$$

$$k_3 = \Delta z \cdot g(r_n + k_2/2), \quad (13)$$

$$k_4 = \Delta z \cdot g(r_n + k_3), \quad (14)$$

$$r_{n+1} = r_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4), \quad (15)$$

where $g(r) \equiv dr/dz$ as defined in (10).

A total of N_{r_0} particles are launched from uniformly spaced initial radial positions $r_0 \in [r_1, r_2]$. A particle is considered lost if it hits the inner or outer electrode wall at any step. The transmission efficiency is computed as the flux-weighted fraction of surviving particles:

$$\Omega_{\text{APM}} = \frac{\sum_{r_0 \in \text{survived}} w(r_0)}{\sum_{r_0 \in \text{all}} w(r_0)}, \quad (16)$$

where the weight $w(r_0)$ accounts for the parabolic flow profile:

$$w(r_0) = \frac{3}{2} \left[1 - \left(\frac{r_0 - r_c}{\delta} \right)^2 \right]. \quad (17)$$

4.3 Cunningham slip correction

The electrical mobility Z_p is related to the mobility diameter D_{mob} via:

$$Z_p = \frac{e C_c(D_{\text{mob}})}{3\pi\eta D_{\text{mob}}}, \quad (18)$$

where η is the dynamic viscosity of air and C_c is the Cunningham slip correction factor (Hinds, 1999):

$$C_c = 1 + \frac{1}{Pd} [15.60 + 7.00 \exp(-0.059 Pd)], \quad (19)$$

with P the atmospheric pressure in kPa and d the diameter in μm .

4.4 Kernel matrix discretisation

For the discrete inversion, the integral in (7) is approximated by the rectangle rule over J mass bins:

$$n(V_i) \approx \sum_{j=1}^J K_{i,j} f(m_j), \quad (20)$$

where $K_{i,j} = \Omega_{\text{APM}}(m_j, V_i, Z_p^*) \cdot \Delta m$ for the 1D model, m_j are uniformly spaced mass grid points, and $\Delta m = (m_{\text{max}} - m_{\text{min}})/(J - 1)$. The kernel matrix $\mathbf{K}$ has shape $I \times J$, where I is the number of voltage bins.

5 Inversion algorithm: Chahine–Twomey method with internal smoothing

5.1 Update equation

To solve the discrete ill-posed problem $\mathbf{n} = \mathbf{K}\mathbf{f}$, we use the non-negative Chahine–Twomey iterative algorithm (Twomey, 1975). The update rule at iteration k is:

$$f_j^{(k+1)} = f_j^{(k)} \cdot \frac{\sum_{i=1}^I \hat{K}_{i,j} \left(n_{\text{meas},i} / n_{\text{calc},i}^{(k)} \right)}{\sum_{i=1}^I \hat{K}_{i,j}}, \quad (21)$$

where $\hat{K}_{i,j} = K_{i,j} / \max(\mathbf{K})$ is the normalised kernel, $n_{\text{meas},i}$ is the measured concentration in the i -th voltage bin, and $n_{\text{calc},i}^{(k)} = \sum_j K_{i,j} f_j^{(k)}$ is the forward-calculated concentration at iteration k .

The initial guess is a uniform distribution:

$$f_j^{(0)} = \frac{\sum_{i=1}^I n_{\text{meas},i}}{m_{\text{max}} - m_{\text{min}}}, \quad j = 1, \dots, J. \quad (22)$$

5.2 Markowski 1-2-1 smoothing

After each iteration, a three-point moving average filter (Markowski, 1987) is applied to suppress high-frequency oscillations caused by noise amplification:

$$\tilde{f}_j = \frac{1}{4} f_{j-1} + \frac{1}{2} f_j + \frac{1}{4} f_{j+1}, \quad j = 2, \dots, J-1. \quad (23)$$

The endpoints ($j = 1$ and $j = J$) are not smoothed.

6 Poisson variance and convergence criterion

6.1 Poisson counting statistics

The CPC measures particle counts N_i during the integration time $t_{\text{meas},i}$ at each voltage bin. The measured concentration is $n_{\text{meas},i} = N_i / V_{\text{sample},i}$, where $V_{\text{sample},i} = Q_{\text{CPC}} \cdot t_{\text{meas},i}$ is the total sampled air volume. Because particle counting follows Poisson statistics, the variance of the measured concentration is:

$$\text{Var}(n_i) = \frac{n_{\text{calc},i}}{V_{\text{sample},i}}. \quad (24)$$

In practice, a floor is applied to prevent division by zero:

$$\text{Var}(n_i) = \max \left(\frac{n_{\text{calc},i}}{V_{\text{sample},i}}, \frac{1}{V_{\text{sample},i}^2} \right). \quad (25)$$

6.2 Reduced chi-squared criterion

Convergence is determined by the reduced chi-squared statistic:

$$\chi^2 = \frac{1}{I} \sum_{i=1}^I \frac{(n_{\text{meas},i} - n_{\text{calc},i})^2}{\text{Var}(n_i)}. \quad (26)$$

The iteration terminates when $\chi^2 < \chi_{\text{threshold}}^2$. The default threshold is $\chi_{\text{threshold}}^2 = 1.0$, corresponding to the expectation that residuals should be comparable to the measurement noise. This criterion prevents overfitting: stopping at $\chi^2 \approx 1$ ensures that the solution explains the data to within the precision allowed by Poisson counting noise.

7 Data preprocessing: voltage binning

7.1 Merging upward and downward scans

The raw APM data consists of time-series measurements during continuous voltage scanning (both upward and downward sweeps). The data are binned into I voltage bins by grouping measurements with similar applied voltages.

7.2 Binning rules for physical quantities

The binning procedure strictly distinguishes between intensive and extensive variables:

- **Intensive variables** (state quantities): Applied voltage V and particle concentration n are averaged within each bin using the arithmetic mean.
- **Extensive variables** (amount quantities): The measurement time t_{meas} for each bin is obtained by summing the individual time intervals Δt of each data row assigned to that bin.

This distinction is critical for the correct evaluation of $V_{\text{sample},i} = Q_{\text{CPC}} \cdot t_{\text{meas},i}$, which directly enters the Poisson variance calculation (25).

8 APM set-point planning with the standalone transfer function

The RK4 trajectory simulation of Sec. 4 can also be used on its own, without measurement data, to inspect the APM transfer function and to plan the operating conditions of an experiment. The tools described in this section do not modify the inversion pipeline.

8.1 Standalone transfer function

For a given rotation speed N (in rpm; $\omega = 2\pi N/60$), mobility diameter D_{mob} and voltage V , the transfer function $\Omega_{\text{APM}}(m; V)$ is evaluated on a mass grid with exactly the same code and coefficients as the 1D kernel, so that $\Omega_{\text{APM}}(m_j; V_i) = K_{i,j}/\Delta m$. The nominal classifying mass, at which centrifugal and electrostatic forces balance at r_c for a singly charged particle, is

$$m_c = \frac{eV}{\omega^2 r_c^2 \ln(r_2/r_1)}. \quad (27)$$

8.2 Resolution parameter and rotation speed

For a spherical particle of diameter d ($= D_{\text{mob}}$) and effective density ρ_{eff} , the mass, mechanical mobility $B = Z_p/e$ and relaxation time are

$$m_p = \frac{\pi}{6} \rho_{\text{eff}} d^3, \quad B = \frac{C_c(d)}{3\pi\eta d}, \quad \tau = m_p B. \quad (28)$$

With the mean axial velocity $\bar{v} = Q/[\pi(r_2^2 - r_1^2)]$, the resolution parameter of Ehara et al. (1996) and the resulting angular velocity are

$$\lambda = \frac{2\tau\omega^2 L}{\bar{v}} \implies \omega = \sqrt{\frac{\lambda\bar{v}}{2\tau L}}. \quad (29)$$

8.3 Voltage matching the transfer-function mode

Equation (10) depends on m and V only through $m\omega^2 r - eV/[r \ln(r_2/r_1)]$. At fixed ω , Z_p and Q , the transfer function is therefore a function of m/V alone, and its mode scales linearly with V . Starting from the force-balance voltage

$$V_0 = \frac{m_p \omega^2 r_c^2 \ln(r_2/r_1)}{e}, \quad (30)$$

the mode m_0^* of $\Omega_{\text{APM}}(m; V_0)$ is located numerically and the voltage is set to

$$V = V_0 \frac{m_p}{m_0^*}, \quad (31)$$

which places the mode exactly at m_p . Because the transfer function has a flat top, the mode is defined as the midpoint of the contiguous region where $\Omega_{\text{APM}} \geq 0.99 \max \Omega_{\text{APM}}$. The transfer function is then re-simulated at (N, V) for verification. In practice V differs from V_0 by less than 10^{-4} (relative) for $\lambda = 0.2$ and by about 0.5% for $\lambda = 0.5$.

Substituting $V \propto m_p \omega^2$ into (10), the right-hand side becomes proportional to $\tau \omega^2 / Q \propto \lambda$ times a function of m/m_p and r . Hence, for a given APM geometry, Ω_{APM} as a function of m/m_p depends only on λ ; e.g. the peak transmission and resolution are 0.847 and $m_p/\text{FWHM} = 2.26$ for $\lambda = 0.2$, and 0.676 and 5.15 for $\lambda = 0.5$ (Kanomax APM-3601 nominal geometry), independent of d , ρ_{eff} and Q . The set-points scale as $N \propto \sqrt{\lambda Q / \tau}$ and $V \propto m_p \omega^2$.

8.4 Multiply charged sphere clusters

Without a DMA upstream of the APM, a randomly oriented cluster of n primary spheres carrying q elementary charges with $n/q = 1$ has the same mass-to-charge ratio as the singly charged monomer and is transmitted at the same set-point. Its mobility is approximated by

$$d_{\text{ve}} = n^{1/3} d, \quad d_{\text{m}} = \chi d_{\text{ve}}, \quad B_{\text{c}} = \frac{C_{\text{c}}(d_{\text{m}})}{3\pi\eta d_{\text{m}}}, \quad (32)$$

with the orientation-averaged dynamic shape factor χ (1.12 for a two-sphere chain; Hinds, 1999, Table 3.2; a continuum-regime value). For mass $M = nm_p$ and charge qe ,

$$\frac{dr}{dz} \propto q B_{\text{c}} \left(\frac{M}{q} \omega^2 r - \frac{eV}{r \ln(r_2/r_1)} \right), \quad (33)$$

so the cluster transfer function is computed with the monomer code using the voltage qV and the mobility diameter d_{m} . As a function of M/q it has the shape of a monomer transfer function with

$$\lambda_{\text{c}} = \frac{2MB_{\text{c}}\omega^2 L}{\bar{v}}. \quad (34)$$

For $n = q = 2$, $\lambda_{\text{c}} \approx 1.26\text{--}1.31 \lambda$ in the range $d = 300\text{--}450$ nm, so the cluster transfer function is narrower and lower than that of the monomer. For monodisperse primaries the relative counting efficiency at a fixed set-point is $\Omega_{\text{c}}/\Omega_1$ evaluated at $m/q = m_p$; the ratio of areas $\int \Omega_{\text{APM}} d(m/q)$ approximates the ratio of voltage-scan-integrated signals. The actual cluster contribution further depends on the cluster concentration and charge distribution.

8.5 Usage

All settings are given in `params.py`:

- **TF*** (standalone transfer function): rotation speed, D_{mob} , list of voltages, and the mass range relative to m_{c} . Run `python run.transfer_function.py`.

- **SP_*** (set-points): list of (d, ρ_{eff}) pairs (nm, kg m⁻³), Q (L/min), λ (a number or a list), mass range relative to m_p , and output directory. **SP_cluster** gives (n, q, χ) of the cluster, or **None** to skip it. Run `python run_apm_setpoint.py`, which writes the transfer functions (CSV), the summary tables `apm_setpoint_summary.csv` and `apm_cluster_summary.csv`, and the figures (for a list of λ , one figure per particle with all λ overlaid).
- `python make_setpoint_report.py` then assembles the conditions, method, tables and figures into a LaTeX/PDF report (`apm_setpoint_report.pdf`) in the output directory.

One set-point requires two transfer-function simulations of 401 mass points (about 40s on a laptop CPU with $N_{r_0} = 1000$ and $\Delta z = 0.1$ mm); a cluster requires one more.

9 Implementation mapping

Table 1 maps each theoretical element to the corresponding function and source file.

Table 1: Mapping from theory to implementation.

Theory	Implementation	File
Cunningham correction, (19)	<code>_cunningham(Dmob)</code>	<code>kernel_simulator.py</code>
Trajectory equation, (10)	<code>dr_dz(rad)</code> inside <code>rk4_transmission</code>	<code>kernel_simulator.py</code>
RK4 integration, (11)–(15)	<code>rk4_transmission(...)</code>	<code>kernel_simulator.py</code>
Transmission efficiency, (16)–(17)	<code>rk4_transmission(...)</code>	<code>kernel_simulator.py</code>
1D kernel matrix, (20)	<code>build_kernel_1d(data, params)</code>	<code>kernel_simulator.py</code>
2D effective kernel, (5)	<code>build_kernel_2d(data, params)</code>	<code>kernel_simulator.py</code>
Chahine–Twomey update, (21)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Markowski smoothing, (23)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Poisson variance, (25)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Chi-squared criterion, (26)	<code>solve_chahine_twomey(...)</code>	<code>inversion_solver.py</code>
Voltage binning, Sec. 7	<code>load_and_bin(params)</code>	<code>data_parser.py</code>
Gaussian mode fitting	<code>fit_gaussian_mode(m_array, f)</code>	<code>visualization.py</code>
Standalone transfer function, (27)	<code>compute_apm_transfer_function(...)</code>	<code>kernel_simulator.py</code>
Set-point, (29)–(31)	<code>find_apm_setpoint(...)</code>	<code>apm_setpoint.py</code>
Cluster, (32)–(34)	<code>simulate_cluster(...)</code>	<code>apm_setpoint.py</code>

10 References

1. K. Ehara, C. Hagwood, and K. J. Coakley, “Novel method to classify aerosol particles according to their mass-to-charge ratio—Aerosol particle mass analyser,” *J. Aerosol Sci.*, vol. 27, no. 2, pp. 217–234, 1996. DOI: 10.1016/0021-8502(96)00014-4.
2. S. Twomey, “Comparison of constrained linear inversion and an iterative nonlinear algorithm applied to the indirect estimation of particle size distributions,” *J. Comput. Phys.*, vol. 18, no. 2, pp. 188–200, 1975. DOI: 10.1016/0021-9991(75)90028-5.
3. G. R. Markowski, “Improving Twomey’s algorithm for inversion of aerosol measurement data,” *Aerosol Sci. Technol.*, vol. 7, no. 2, pp. 127–141, 1987. DOI: 10.1080/02786828708959153.
4. W. C. Hinds, *Aerosol Technology: Properties, Behavior, and Measurement of Airborne Particles*, 2nd ed. Wiley, 1999. Eq. (3.22) and Table 3.2.

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